

Generative AI for Business Analysis and Product Design



Estimated time: 40 minutes

Welcome to the project - Generative AI for business analysis and product design

Learning objectives

After completing this project, you will be able to:

- Demonstrate examples of leveraging generative AI for market research, competitor analysis, and pricing strategy
- Demonstrate examples of leveraging generative AI for idea generation and brainstorming.
- Demonstrate examples of content creation for marketing through generative AI.
- Identify how generative AI can be useful for personalized product recommendations for customers.

Introduction

The various applications of generative AI offer opportunities for business analysis and increased productivity across different domains within an organization. This is particularly significant for emerging startups or for established organizations planning to expand their business and operations.

Generative AI offers exciting opportunities and benefits to gain a competitive edge and accelerate growth:

- Discovering valuable insights into market trends, consumer preferences, and competitor analysis.
- Obtaining recommendations about operational dynamics such as pricing strategies, marketing strategies, and brand positioning.
- Boosting efficiency and productivity by automating repetitive tasks, enabling high-quality content creation for marketing, and implementing AI-powered chatbots for handling simple customer requests.
- Enhancing creativity and innovation through idea generation and brainstorming, implementing personalized marketing campaigns, and getting insights into customer preferences and market trends by analyzing datasets of social media.
- Improving operational efficiency through fraud detection, predictive maintenance, managing inventory levels, and predicting demand.

Scenario

In this project, you will work on use cases to demonstrate the use of generative AI for gathering insights and generating ideas to launch a new product successfully. These use cases will be useful to senior management for strategic alignment and decision-making. They also benefit professionals in various domains, including marketing research and product design and development.

For this project, let's consider a well-established brand, 'Pure Glow,' renowned for its premium skincare and cosmetics products. The brand has decided to expand its product line to include body cleansers. With a loyal customer base and a reputation for quality and innovation in the skincare industry, Pure Glow aims to leverage its expertise and brand equity to capture a larger share of the market.

The company aims to offer gentle and effective cleansers that cleanse, hydrate, and nourish the skin. As part of its expansion strategy, Pure Glow plans to introduce body cleansers formulated with natural ingredients, catering to various skin types and preferences.

In such a scenario, Pure Glow can leverage generative AI for various business and product aspects, including market research, competitor analysis, pricing strategy, product labeling, social media campaigns, personalized promotions, and so on.

Generative AI tool used in the project:

The examples in this project are demonstrated using the Generative AI Classroom.

The Generative AI Classroom is a tool where you can write and compare your prompts to generate desired text with real-time chat responses.

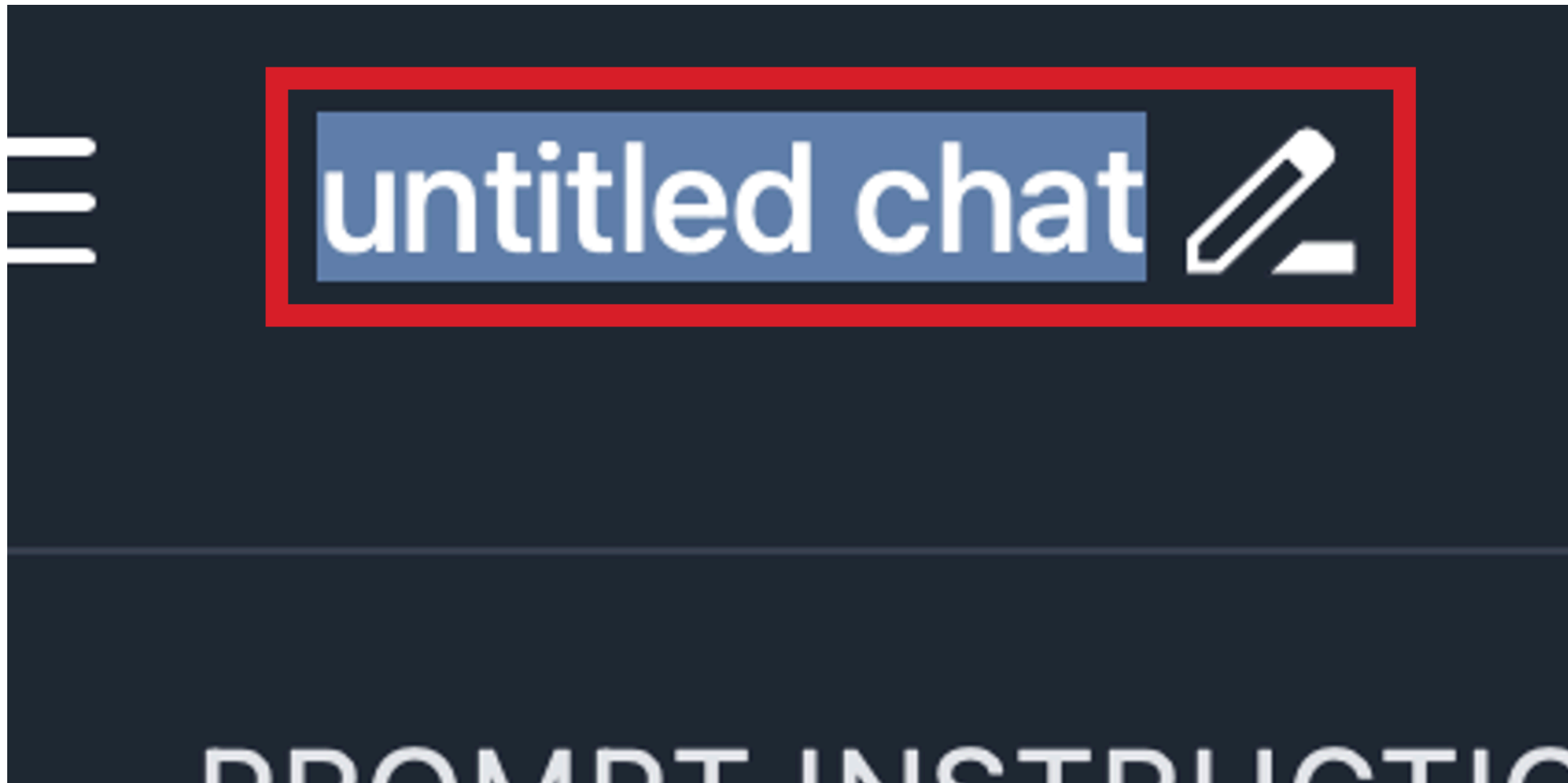
Exercise 1: Setting up the environment

As a first step, you must set up your AI classroom for a better learning experience.

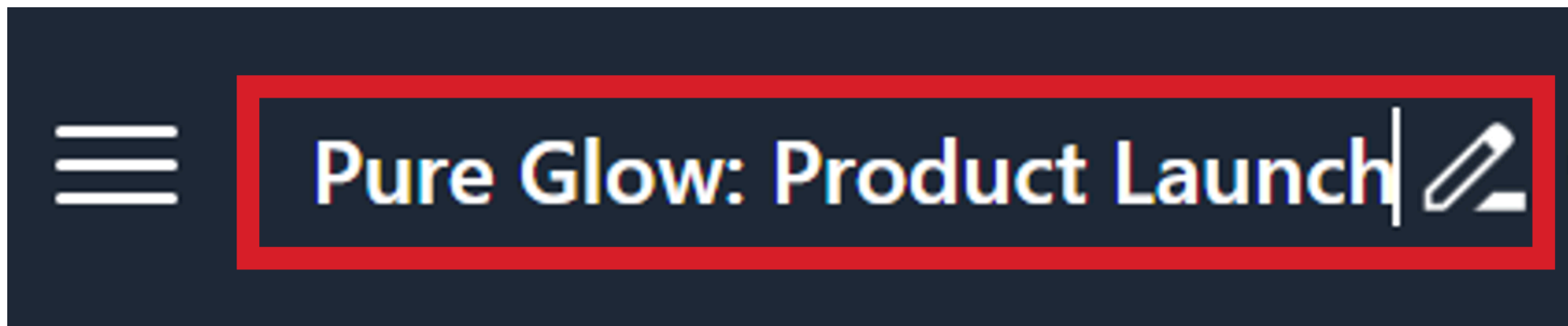
Before beginning the exercise, you must:

1. Name the chat:

Use the pencil icon available on the top-left corner of the right pane to name the chat.

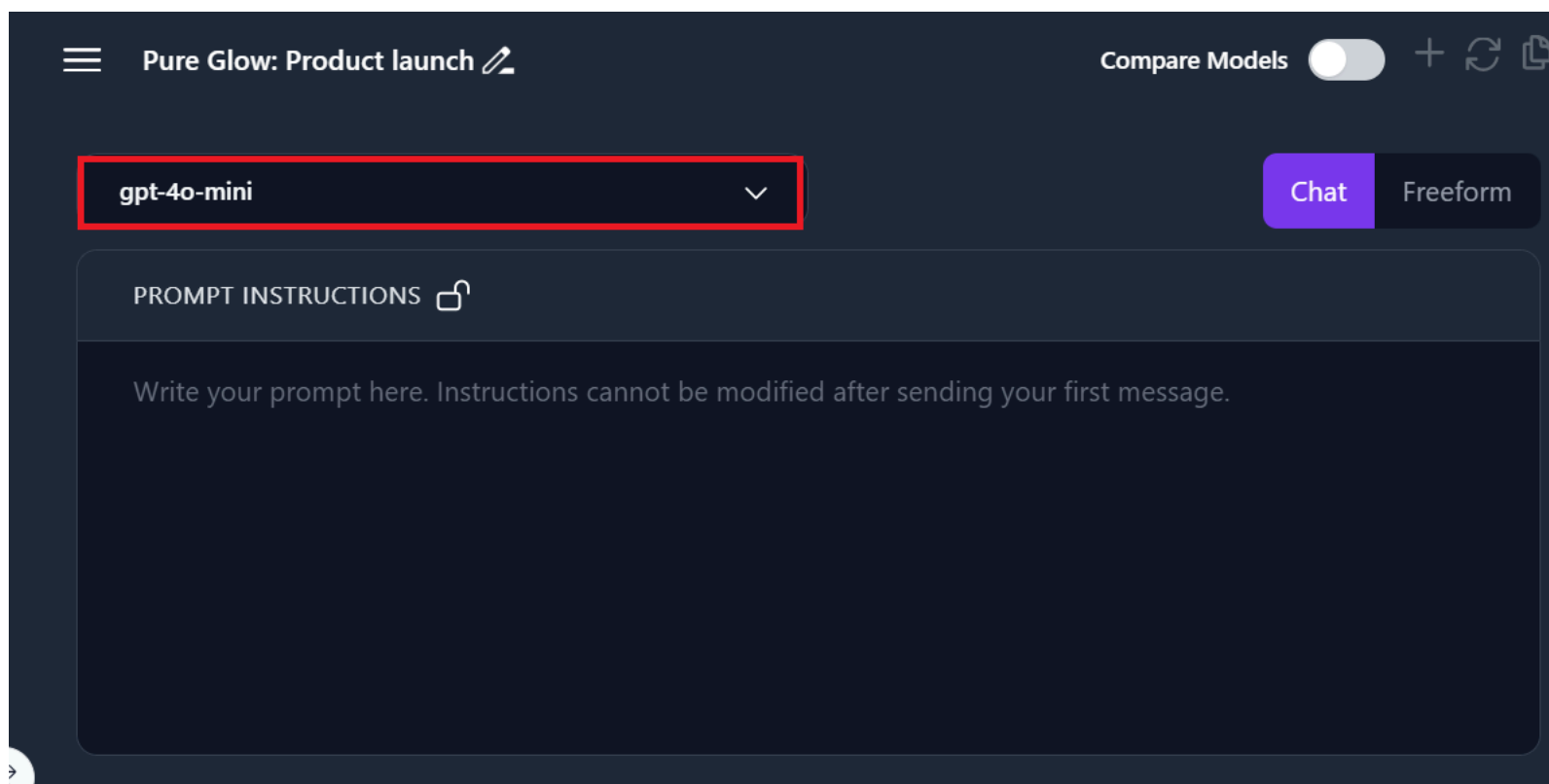


For our scenario, let us name the chat 'Pure Glow: Product Launch'.



2. Choose the model:

Use the dropdown option from the top-right corner of the right pane for the text generation exercise.



3. Give prompt instructions:

Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.

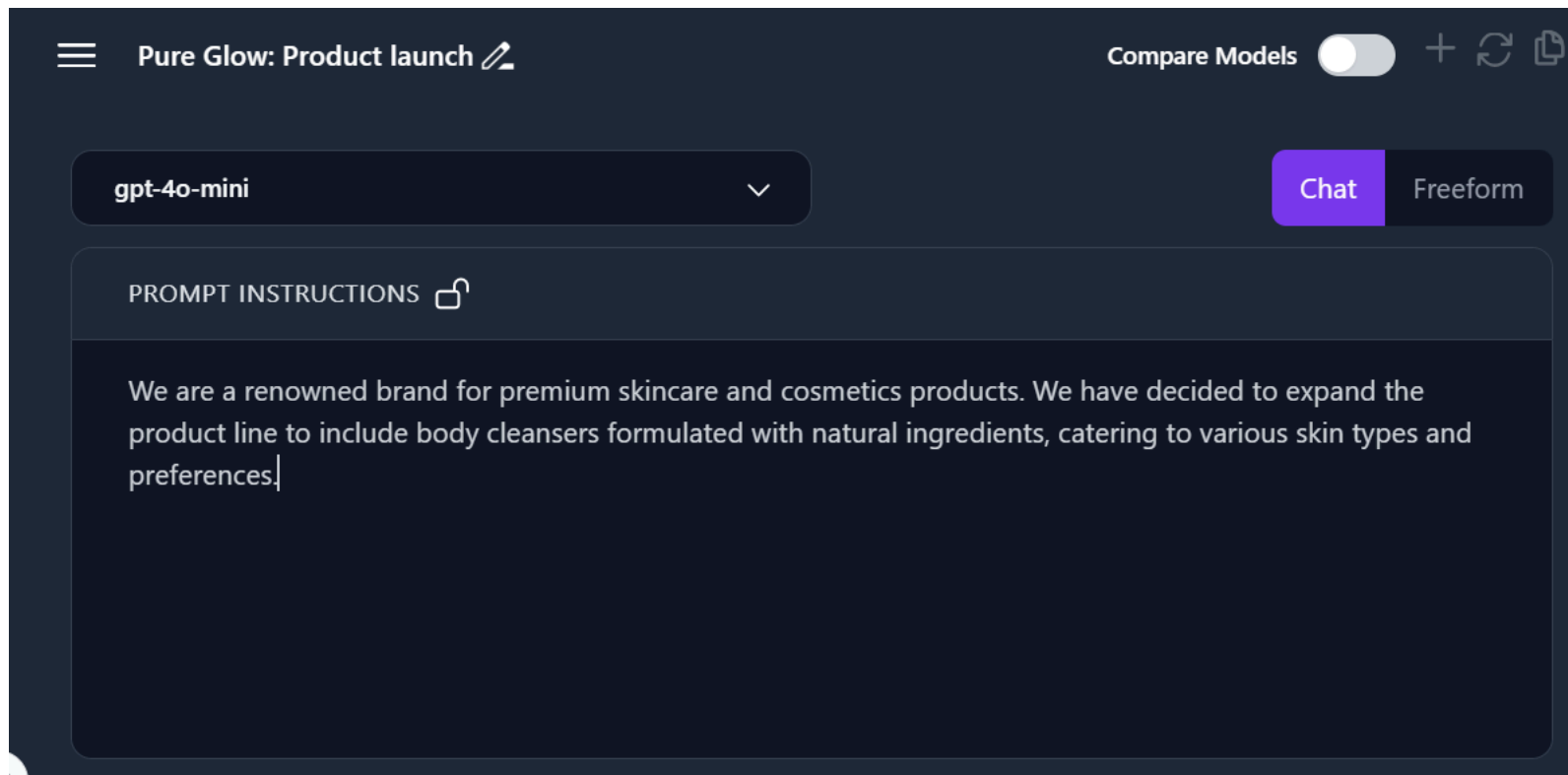
PROMPT INSTRUCTIONS

Write your prompt here. Instructions can



You should have clear and specific prompt instructions, such as:

We are a renowned brand for premium skincare and cosmetics products. We have decided to expand the product line to include body cleansers formulated with natural ingredients, catering to various skin



Now, we are all set to discover different facets of the business and product within this context. Let's first consider the use cases through the lens of Pure Glow's business objectives.

Exercise 2: Leveraging generative AI for business analysis

With this, let's understand how Pure Glow can leverage the use of generative AI for business analysis to enhance its strategic decision-making and improve operational efficiency in the competitive body cleanser market.

Example 1: Market Research

For a new product, it is important to first analyze the market trends and consumer preferences. In this example, we will demonstrate how generative AI can help to provide valuable insights for making informed decisions and developing competitive strategies.

1. Once we have provided the prompt instructions, we can start a chat to gain the required information. Let's enter the following message in the **Start chat** field.

Identify the trends in the market for body cleanser products.

Identify the trends in the market for body cleanser products.

Start chat

powered by **watsonx**[™]

2. Click the **Start chat** icon to generate the output.

gpt-4o-mini



Chat

Freeform

PROMPT INSTRUCTIONS 📄

We are a renowned brand for premium skincare and cosmetics products. We have decided to expand the product line to include body cleansers formulated with natural ingredients, catering to various skin types and preferences.



How to use AI Classroom

1

Name the chat

2

Choose your model(s). You may select multiple models when "Compare Models" toggle is active

3

Set the prompt instructions

4

Chat with your language model

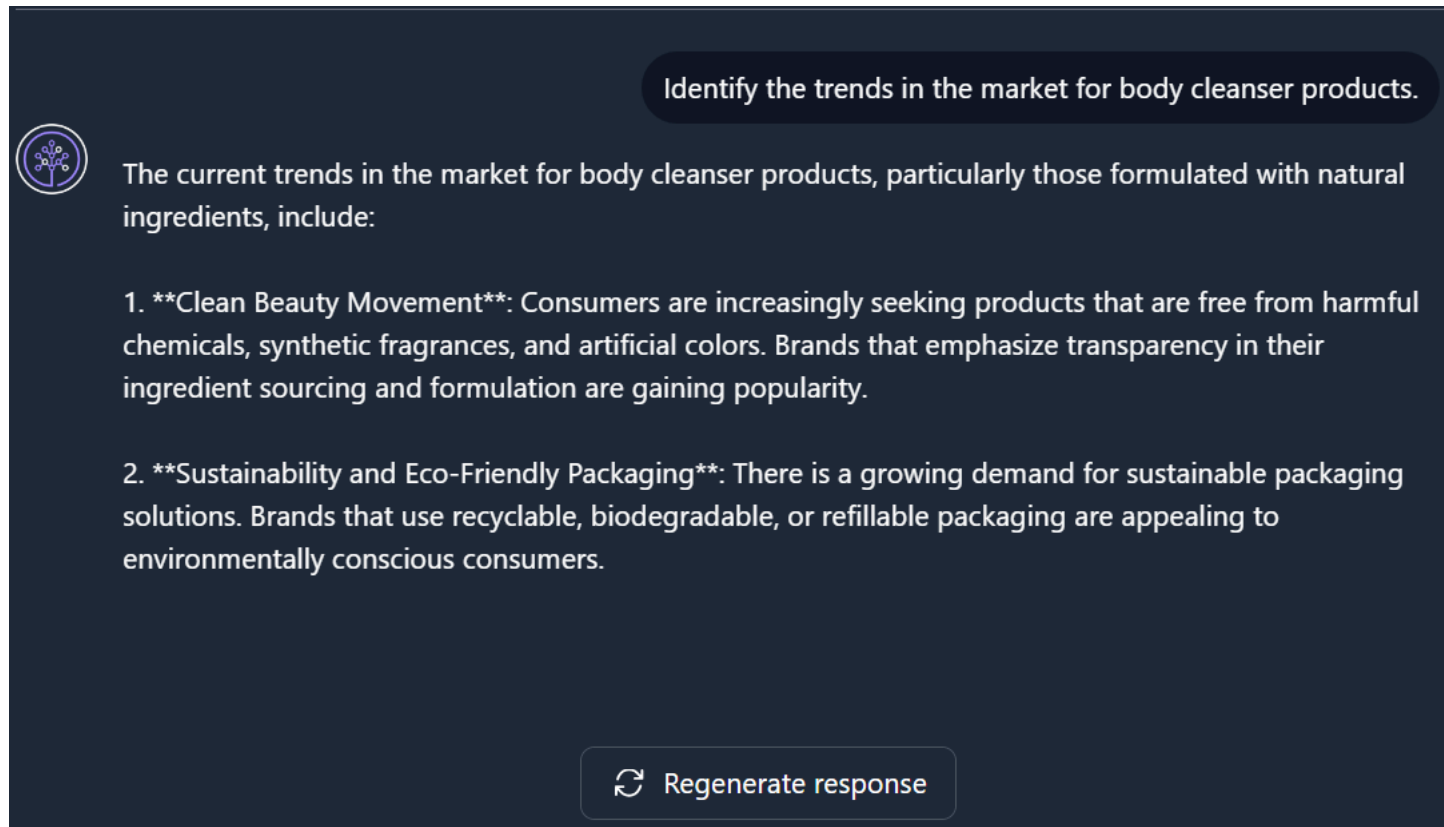


Want to elevate your AI learning experience? Check out [IBM watsonx.ai](https://ibm.com/watsonx.ai) now!

Identify the trends in the market for body cleanser products.

Start chat

3. We received the output shown in the following image.



The screenshot shows a chat interface with a dark background. At the top, a prompt in a rounded rectangle says "Identify the trends in the market for body cleanser products." Below this, on the left, is a circular icon with a stylized plant. To the right of the icon, the response text reads: "The current trends in the market for body cleanser products, particularly those formulated with natural ingredients, include:" followed by two numbered points. The first point is about the "Clean Beauty Movement" and the second is about "Sustainability and Eco-Friendly Packaging". At the bottom right of the chat area is a button with a circular arrow icon and the text "Regenerate response".

Identify the trends in the market for body cleanser products.

The current trends in the market for body cleanser products, particularly those formulated with natural ingredients, include:

1. **Clean Beauty Movement**: Consumers are increasingly seeking products that are free from harmful chemicals, synthetic fragrances, and artificial colors. Brands that emphasize transparency in their ingredient sourcing and formulation are gaining popularity.
2. **Sustainability and Eco-Friendly Packaging**: There is a growing demand for sustainable packaging solutions. Brands that use recyclable, biodegradable, or refillable packaging are appealing to environmentally conscious consumers.

Regenerate response

► [Click here for complete output](#)

Try Yourself!

Feel free to continue exploring market trends by asking more questions to the tool. This will give you a deeper insight into making informed business decisions. Try to design the prompts regarding the following aspects and generate the response from the tool.

1. Potential market demographics

► [Click here for an example of a prompt](#)

2. Potential gap in this market segment

► [Click here for an example of a prompt](#)

Example 2: Competitor analysis

A thorough competitor analysis can help businesses gain valuable insights into the competitive landscape and successfully launch their product in the market.

1. To understand the current competition in the market, let's continue our current chat and enter the following message:

Provide insights into our competitor's strengths and weaknesses.



Provide insights into our competitor'

2. Click the **Send** icon.

Provide insights into our competitor'

3. We received the output shown in the following image.



Provide insights into our competitor's strengths and weaknesses.

To provide insights into your competitors' strengths and weaknesses in the body cleanser market, it's essential to analyze key players in the industry. Here are some general strengths and weaknesses commonly observed among competitors in the premium skincare and natural body cleanser segment:

Competitor Strengths:

1. ****Established Brand Reputation****: Many competitors have built strong brand loyalty and recognition over the years, which can create trust and credibility among consumers.
2. ****Diverse Product Offerings****: Competitors often have a wide range of products catering to different skin types and concerns, allowing them to appeal to a broader audience.

► [Click here for complete output](#)

4. You may also regenerate the response. This feature allows you to generate a new response based on the current conversation context.

To do so, select the **Regenerate response** option.

brand and attract customers looking for

Type your message

5. The regenerated output is shown.



To effectively analyze your competitors in the body cleanser market, it's essential to evaluate their strengths and weaknesses. Here are some insights that can guide your understanding:

Competitor Strengths:

1. ****Established Brand Loyalty****: Many competitors may have a loyal customer base built over years of quality products and effective marketing. This loyalty can make it challenging for new entrants to gain market share.
2. ****Diverse Product Range****: Competitors often offer a wide variety of body cleansers, catering to different skin types and preferences. This diversity can attract a broader audience and meet various consumer needs.

 Regenerate response

► [Click here for Complete output](#)

Try Yourself!

Try experimenting to obtain more information for competitor analysis. Design the prompts regarding the following aspects and generate the response from the tool.

1. Opportunities to capitalize on competitors' weaknesses

► [Click here for an example of a prompt](#)

2. Unique features to differentiate the product from competitors

► [Click here for an example of a prompt](#)

Example 3: Pricing strategy

Developing a pricing strategy for your body cleanser product is crucial for achieving profitability while remaining competitive in the market. Let's see how generative AI can help you to formulate an effective pricing strategy.

1. To ask a question about the pricing strategy, let's enter the following prompt:

What pricing strategy would be most competitive yet profitable for my body cleanser product?

What pricing strategy would be most

2. Select **Send** to receive the response.

What pricing strategy would be most

3. We received the following output.

What pricing strategy would be most competitive yet profitable for my body cleanser product?



To establish a competitive yet profitable pricing strategy for your body cleanser products, consider the following approaches:

1. **Value-Based Pricing**:

- **Definition**: Set prices based on the perceived value of your product to the consumer rather than solely on production costs.
- **Implementation**: Conduct market research to understand what features and benefits consumers value most (e.g., natural ingredients, sustainability, effectiveness). Price your products to reflect this perceived value, ensuring customers feel they are getting a premium product for a fair price.

2. **Competitive Pricing**:



Regenerate response

► [Click here for complete output](#)

Try Yourself!

You can also discover other aspects, such as the operational efficiency of the business, or scaling and growth of the product by asking questions, such as:

1. Operational efficiency

► [Click here for an example of a prompt](#)

2. Market expansion

► [Click here for an example of a prompt](#)

Exercise 3: Product design

Let us also look at the usage of generative AI for product design. We will consider various product-specific use cases and understand how generative AI can be leveraged for the same.

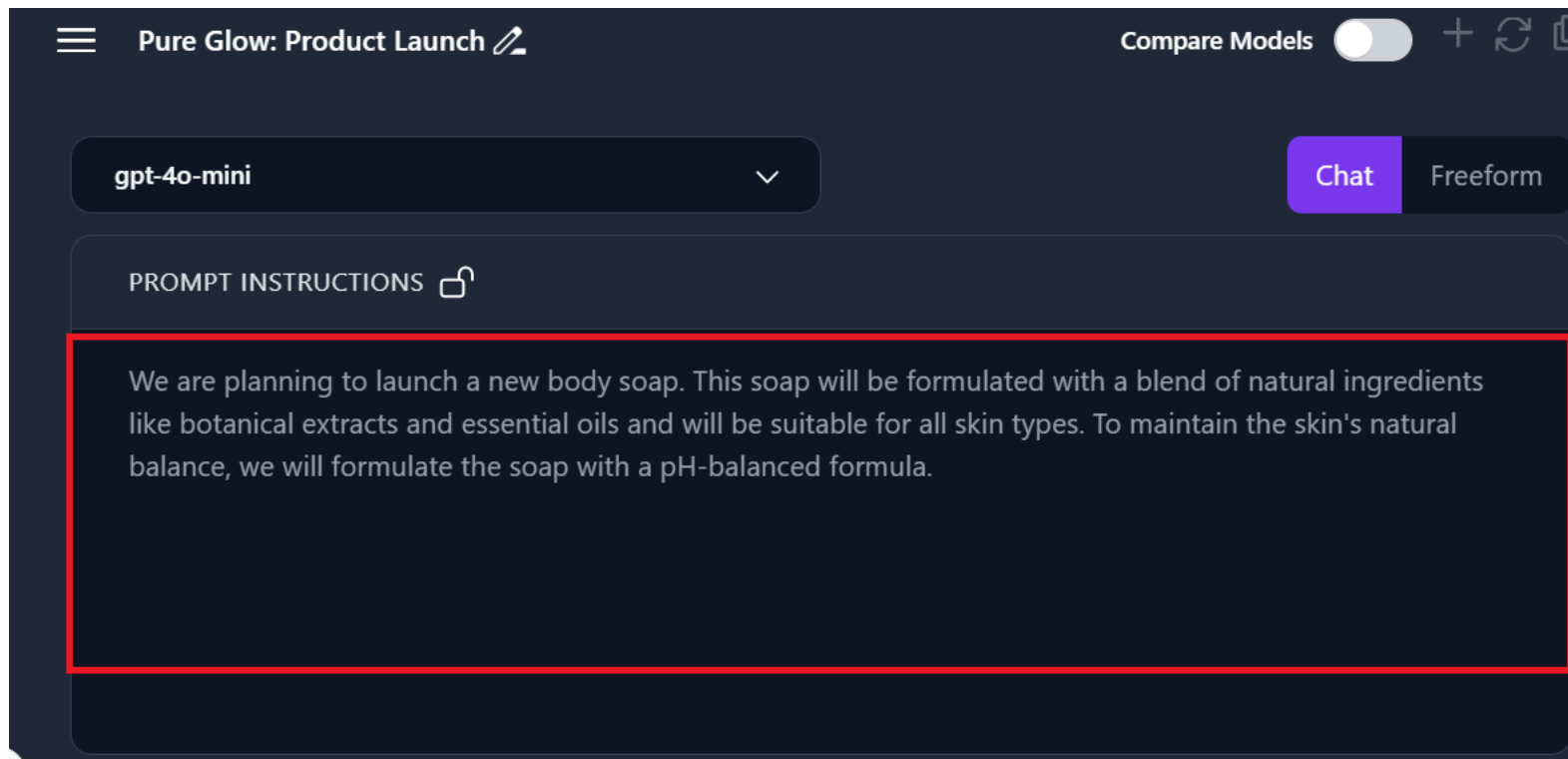
Scenario

As part of the new offering for body cleansers, Pure Glow plans to launch a new body soap. This soap will be formulated with a blend of natural ingredients like botanical extracts and essential oils and will be suitable for all skin types. To maintain the skin's natural balance, we will formulate the soap with a pH-balanced formula.

Example 1: Generate the product name

1. Let's enter the following instructions in the Prompt Instructions field to set the context, highlighting the characteristics or any other relevant information about the soap.

We are planning to launch a new body soap. This soap will be formulated with a blend of natural ingredients like botanical extracts and essential oils and will be suitable for all skin types. To main



2. To generate the product name, let's enter the following prompt in the **Type your message** field in the same chat.

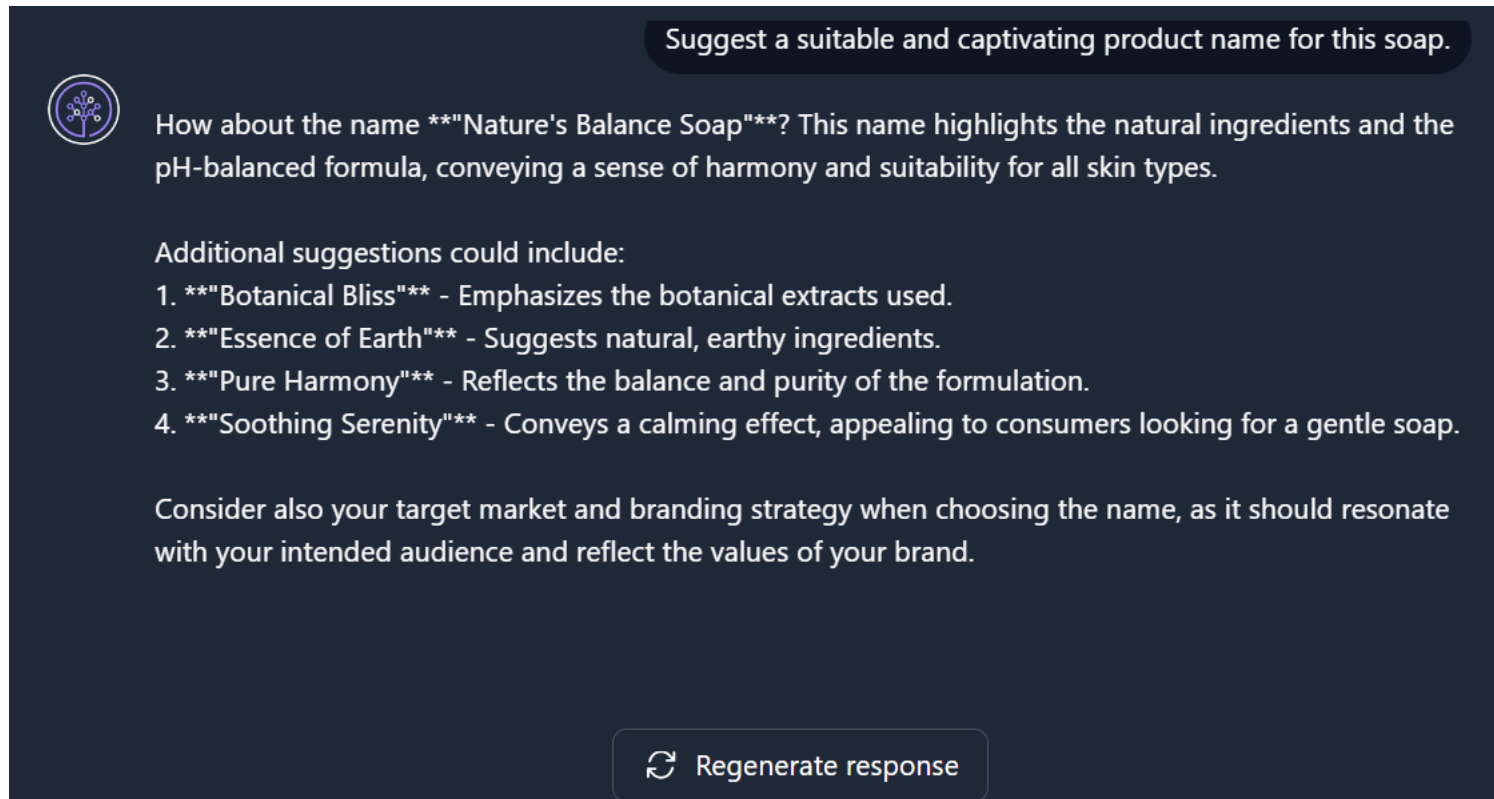
Suggest a suitable and captivating product name for this soap.

Suggest a suitable and captivating product name

3. Click the **Send** icon to generate the output.

Suggest a suitable and captivating product name

4. We received the output shown in the following image.



5. Let us continue the conversation to refine or modify the generated name. For example, in this output, we find that the generated product name is a little long and we would prefer a shorter name. Let's enter the following prompt.

Modify this product name to reduce words or suggest an alternative, shorter product name.



Modify this product name to reduce words or suggest an alternative, shorter product name.

How about the name **"Balance Soap"**? This concise name still conveys the essence of your product's pH-balanced formula while being easy to remember.

Additional shorter alternatives could include:

1. **"Botanica"** - A catchy, memorable name that suggests botanical ingredients.
2. **"Pure Soap"** - Simple and effective, highlighting the purity of the formulation.
3. **"Serene Soap"** - Evokes a sense of calm and relaxation.
4. **"Essence Soap"** - Suggests a focus on natural, essential ingredients.

These options maintain the appeal of your product while being more succinct.

6. The output produced at our end is shown in the image. We preferred to work forward with the name 'Botanica'.

Example 2: Generate the tagline

1. Next, let's generate a tagline supporting this product name. Let's enter the following prompt.

Considering the product name 'Botanica' for this soap, suggest a tagline that reinforces the product identity and resonates with the target audience.

Considering the product name 'Botanica' for this soap, suggest a tagline that reinforces the product identity and resonates with the target audience.

Send

2. Click the **Send** icon to generate the output. The output produced is shown in the image.



A suitable tagline for the product name **Botanica** could be **"Embrace Nature, Elevate Your Skin."** This tagline reinforces the natural essence of the product while highlighting its benefits for skin health.

Additional tagline suggestions include:

1. **"Pure Ingredients, Pure Indulgence."** - Emphasizing the quality and luxurious feel of the soap.
2. **"Nature's Best for Every Body."** - Highlighting the inclusivity of the product for all skin types.
3. **"Revitalize with Botanical Goodness."** - Focusing on the rejuvenating properties of the botanical extracts.
4. **"Cleanse, Nourish, Flourish."** - Suggesting a holistic approach to skincare with the soap.

When selecting a tagline, ensure it aligns with your brand's mission and resonates with consumers who value natural and effective skincare products.

Try Yourself!

Try experimenting with the generative AI tool to ask further questions about product design. Design the prompts regarding the following aspects and generate the response from the tool.

1. Design concepts for offering soap as a bar as well as a dispenser

► [Click here for an example of a prompt](#)

2. Eco-friendly packaging

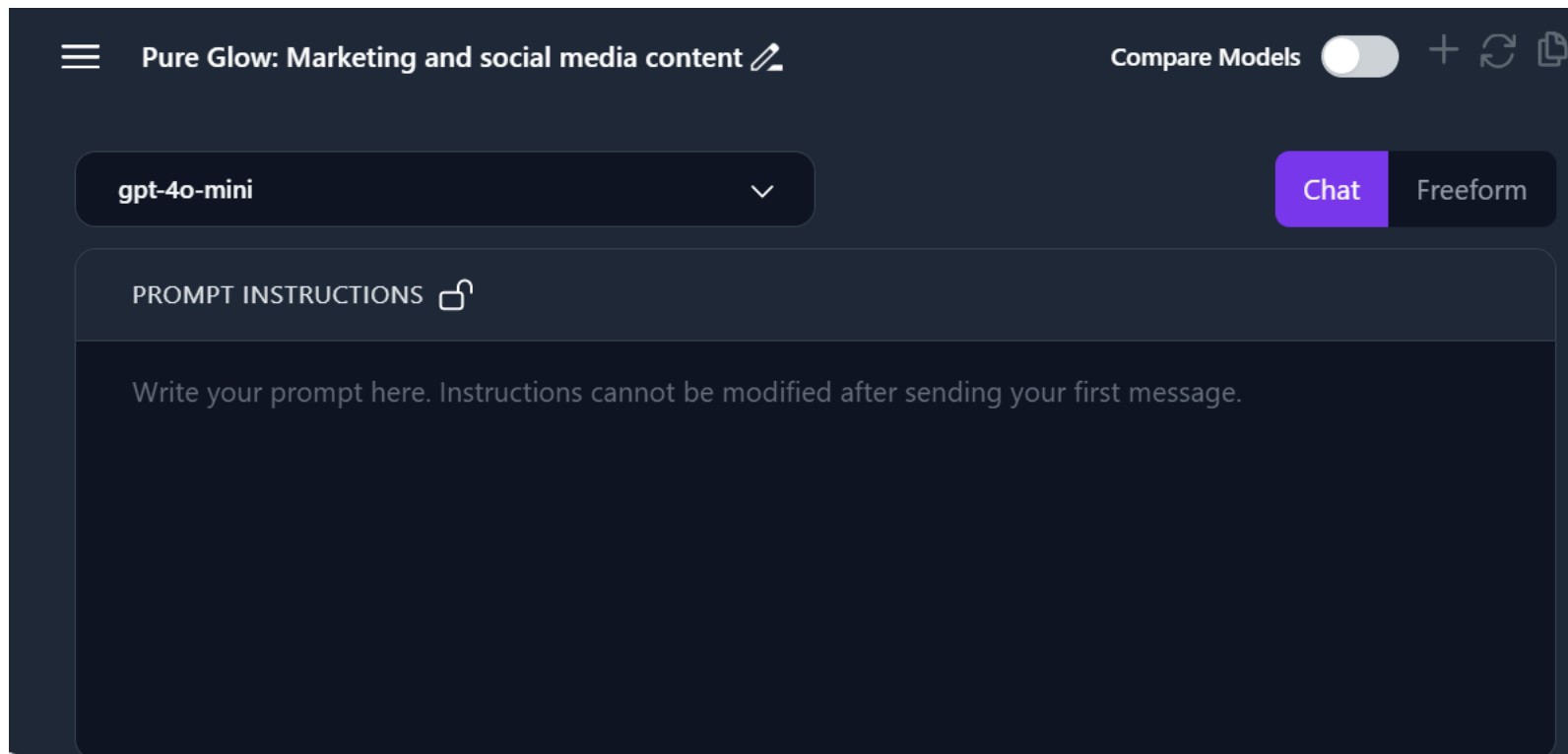
► [Click here for an example of a prompt](#)

Exercise 4: Content creation for marketing and social media

For the launch of a new product, generative AI can be leveraged to generate content for marketing and social media, including marketing copy, captions, emails, product descriptions, and blog posts.

In this exercise, we will demonstrate some of these use cases for the fictitious organization Pure Glow, which is planning to launch its new product, a body soap named 'Harmony Wash'. We will experiment with generative AI to generate captions, posts, and social media artifacts.

1. Ensure that you have set up your AI classroom by providing the name to the chat and setting the appropriate model.



2. Draft the prompt instructions, including relevant details. Let's enter the following prompt instructions.

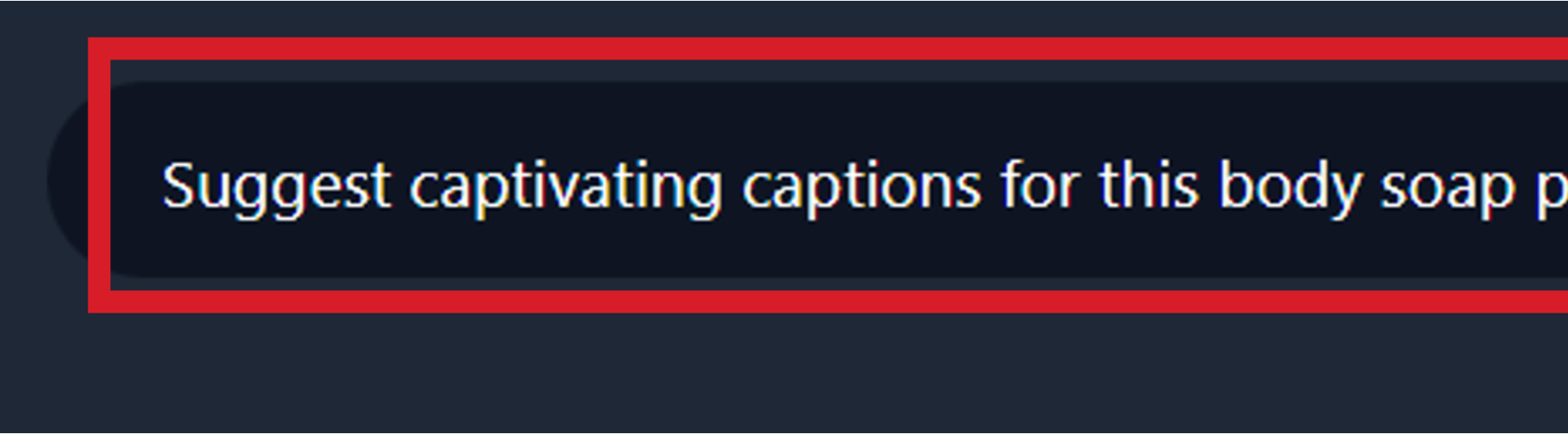
We are planning to launch a new body soap, named Harmony Wash. This soap will be formulated with a blend of natural ingredients like botanical extracts and essential oils and will be suitable for all

PROMPT INSTRUCTIONS

We are planning to launch a new body soap, named "Santal de Indes", which contains sandalwood extracts and essential oils and will be suitable for

3. Next, in the **Type your message** field, enter and send the following:

Suggest captivating captions for this body soap promoting freshness and wellness.



Suggest captivating captions for this body soap p

4. In response to the prompt, the following output is generated displaying a number of captions.

Suggest captivating captions for this body soap promoting freshness and wellness.

Here are some captivating captions for your Harmony Wash body soap that promote freshness and wellness:

1. "Awaken your senses with Harmony Wash – where nature meets nourishment! 🌿 ✨"
2. "Embrace the freshness of nature with every wash. Your skin deserves Harmony! 💧 🌸"
3. "Refresh, rejuvenate, and restore with our all-natural Harmony Wash. Feel the difference! 🌼 🌿"
4. "Dive into a world of botanical bliss. Harmony Wash – your daily dose of freshness! 🌊 🌿"
5. "Elevate your self-care routine with Harmony Wash. Pure ingredients for pure wellness! ✨ 🛀"
6. "Nourish your skin and soul with the harmonious blend of nature's best. ❤️ 🌱"
7. "Freshness is just a wash away! Discover the power of nature with Harmony Wash. ☀️ 🍋"
8. "Suds of serenity await! Experience the refreshing embrace of Harmony Wash. 🌈 ✨"
9. "Find your balance with every lather. Harmony Wash – the essence of wellness! 🌿 🌊"
10. "Nature's touch, bottled for your bliss. Feel fresh, feel alive with Harmony Wash! 🌸 🌍"

These captions not only highlight the freshness and wellness aspects of your product but also emphasize its natural ingredients, appealing to eco-conscious consumers.

5. Let's continue the conversation to experiment with generating marketing content from different viewpoints or styles to cater to varied audiences. Let's create a social media post targeting individuals conscious of their skin's well-being. In the same session, enter the following prompt.

Create a social media post targeting individuals who are mindful and caring about their skin's well-being, emphasizing how this soap is suitable for all skin types.

Create a social media post targeting individuals w
for all skin types.

6. The response generated at our end is displayed here.

Create a social media post targeting individuals who are mindful and caring about their skin's well-being, emphasizing how this soap is suitable for all skin types.



🌿 ✨ ****Introducing Harmony Wash: Your Skin's Best Friend!**** ✨ 🌿

At Harmony Wash, we believe that every skin type deserves the gentle care of nature. Our body soap is thoughtfully formulated with a blend of botanical extracts and essential oils, ensuring a refreshing and nourishing experience for everyone!

💧 ****Why Choose Harmony Wash?****

- ****All-Natural Ingredients:**** Embrace the power of nature with every wash.
- ****Suitable for All Skin Types:**** Whether you have dry, oily, or sensitive skin, Harmony Wash is here to cater to your unique needs.
- ****Mindfully Crafted:**** We prioritize your skin's health and the planet's well-being, using sustainable practices in our production.

🌸 ****Join the Harmony Movement!****

Feel good about what you put on your skin. Treat yourself to a wash that not only cleanses but also promotes your overall wellness.

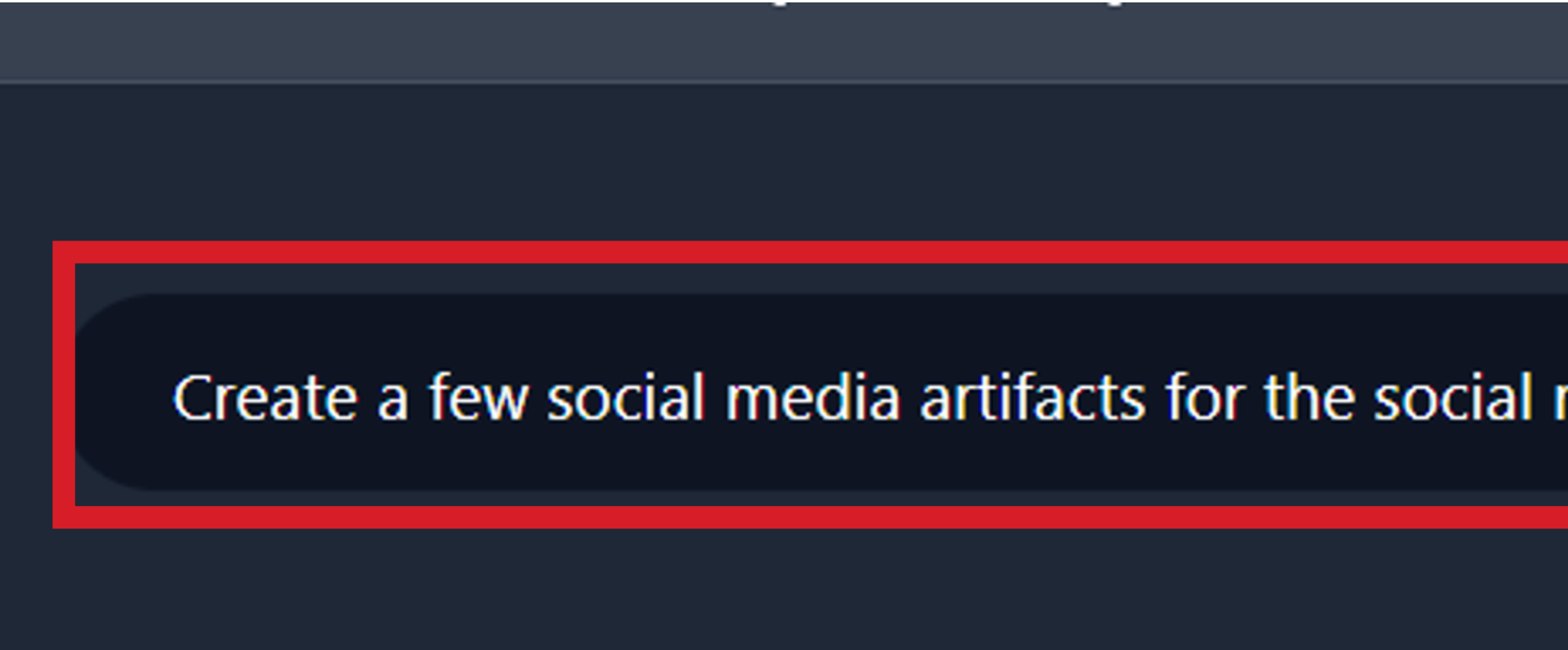
✨ ****Tag a friend who deserves to experience the harmony of nature!****

#HarmonyWash #NaturalBeauty #SkinWellness #EcoFriendly #AllSkinTypes #BotanicalBliss
#SelfCareJourney

This post resonates with individuals who are mindful about their skin and emphasizes the soap's suitability for all skin types, while also promoting a sense of community and care.

7. Next, let's create a few social media artifacts for the social media campaign for this soap. In the same session, let's enter the following prompt.

Create a few social media artifacts for the social media campaign for this soap.



Create a few social media artifacts for the social r

8. The output produced at our end is shown here.

Here are a few social media artifacts you can use for your Harmony Wash campaign, including graphics ideas, hashtags, and post formats:

1. **Instagram Carousel Post**

- **Slide 1:** Eye-catching image of Harmony Wash with natural ingredients in the background (e.g., botanical extracts, essential oils).

- Caption: "Meet Harmony Wash: Your skin's new best friend! 🌿✨"

- **Slide 2:** Infographic highlighting the benefits of Harmony Wash (all-natural, suitable for all skin types, cruelty-free).

- Caption: "Crafted with care, made for everyone. Discover the harmony of nature! 💚💧"

- **Slide 3:** Customer testimonial or quote about the soap's refreshing feel.

- Caption: "What our users are saying: 'My skin feels revitalized and so soft!' - @happycustomer 🌸"

2. **Instagram Stories**

- **Story 1:** Short video clip of the soap lathering in water, showcasing its texture and bubbles.

- Text Overlay: "Lather up with nature! 🛁✨"

- **Story 2:** Poll asking followers, "What's your skin type?" with options: "Dry" / "Oily" / "Sensitive" / "Combination."

- **Story 3:** Countdown sticker for the launch date of Harmony Wash.

- Text Overlay: "Get ready for the ultimate skin refresh! Launching in [X days]! 🚀"

3. **Facebook Post**

- **Graphic:** A serene image of a person enjoying a shower with Harmony Wash, surrounded by plants.

- Caption: "Indulge in the essence of wellness with Harmony Wash! 🌸✨ Suitable for all skin types, our all-natural formula will leave your skin feeling fresh and nourished. Are you ready to embrace the harmony? 🌿💖 #HarmonyWash #SkinCare #WellnessJourney"

4. **Twitter Post**

- **Tweet:** "Say goodbye to harsh soaps! ✨ Harmony Wash is here to pamper your skin with the gentle touch of nature. Suitable for all skin types! 💧🌸 #HarmonyWash #NaturalBeauty"

These artifacts can help create a visually appealing and engaging social media campaign that resonates with your target audience and promotes the unique qualities of Harmony Wash.

Try Yourself!

Try experimenting with the generative AI tool to ask further questions about marketing and promotion of the product. Design the prompts regarding the following aspects and generate the response from the tool.

1. Promotional strategies

► [Click here for an example of a prompt](#)

2. Brand narrative

► [Click here for an example of a prompt](#)

Exercise 5: Personalized product recommendations for customers

Generative AI can help organizations promote their products by providing personalized recommendations to customers based on their individual characteristics and past purchases.

To demonstrate this, we will consider data about a few fictitious customers and fictitious products offered by 'Pure Glow'. As Pure Glow specializes in body cleansers and cosmetic products, we will prompt Gen AI with a few cleansing products and their descriptions and data about the skin types, preferences, and purchase histories of a few customers. Based on the prompted data, we will generate personalized product recommendations for these customers.

1. Ensure that you have set up your AI classroom by providing the name to the chat and setting the appropriate model.

The screenshot shows the 'Pure Glow: Personalized product recommendation' chat interface. At the top, there is a header bar with a hamburger menu icon on the left, the title 'Pure Glow: Personalized product recommendation' with an edit icon, and a 'Compare Models' toggle switch on the right. Below the header, there is a model selection dropdown menu currently set to 'gpt-4o-mini'. To the right of the model selection are two buttons: 'Chat' (highlighted in purple) and 'Freeform'. Below these is a section titled 'PROMPT INSTRUCTIONS' with a lock icon. The main area of the interface is a large text box containing the instruction: 'Write your prompt here. Instructions cannot be modified after sending your first message.'

2. Draft the prompt instructions, including relevant details. Let's enter the following prompt instructions.

Pure Glow specializes in body cleansers and cosmetic products. They want to promote the following products offered by them:

- SmartCleanse Body Wash: A gentle, hydrating body wash enriched with natural extracts.
- Radiance Scrub Gel: A rejuvenating body scrub with vitamin-infused gel.
- LightGlow Body Lotion: A lightweight, non-greasy lotion designed to balance the skin's natural moisture levels.
- GlowGuard Sunscreen Mist: Broad-spectrum SPF mist for sun protection without a heavy feel.

Here is some data about a few customers:

Name: Emily Johnson

Skin Type: Oily

Preferences: prefers cruelty-free products

Purchase History: Bought a floral-scented cleanser and a cruelty-free moisturizer in the past

Name: Michael Smith

Skin Type: Sensitive

Preferences: Prefers unscented and hypoallergenic products

Purchase History: Bought an unscented hypoallergenic body wash and a skin lotion in the past

Name: Jessica Martinez

Skin Type: Combination

Preferences: Enjoys citrus scents, prefers organic and natural ingredients

Purchase History: Bought a citrus-scented body scrub and an organic facial cleanser in the past

☰

Pure Glow: Personalized product recommendation 

Compare Models ☐ + ↺ 📄

gpt-4o-mini ▾

Chat Freeform

PROMPT INSTRUCTIONS 

Pure Glow specializes in body cleansers and cosmetic products. They want to promote the following products offered by them:

- SmartCleanse Body Wash: A gentle, hydrating body wash enriched with natural extracts.
- Radiance Scrub Gel: A rejuvenating body scrub with vitamin-infused gel.
- LightGlow Body Lotion: A lightweight, non-greasy lotion designed to balance the skin's natural moisture levels.
- GlowGuard Sunscreen Mist: Broad-spectrum SPF mist for sun protection without a heavy feel.

3. Next, in the **Type your message** field, enter and send the following:

Generate personalized product recommendations for each customer based on their individual characteristics and past purchases.

How to

1



Want to elevate your AI le

Generate personalized product recommendations based on user characteristics and past purchases. |

3. Click the **Start chat** icon. Based on the prompt, we received the following output.



Here are personalized product recommendations for each customer based on their skin type, preferences, and purchase history:

Emily Johnson

****Skin Type:**** Oily

****Preferences:**** Prefers cruelty-free products

****Purchase History:**** Bought a floral-scented cleanser and a cruelty-free moisturizer

****Recommended Products:****

1. ****SmartCleanse Body Wash**** - This gentle, hydrating body wash is enriched with natural extracts, making it suitable for Emily's oily skin while aligning with her preference for cruelty-free products.
2. ****Radiance Scrub Gel**** - Although she hasn't purchased a scrub in the past, this rejuvenating body scrub with vitamin-infused gel can help balance her skin and remove excess oil without being harsh.

► [Click here for complete output](#)

Summary

Congratulations! You just completed the project Generative AI for business analysis and product design.

In this project, you explored a few use cases of generative AI for organizations or for launching a new product. In this project, you demonstrated generating the market research, competitor analysis, product name, and tagline for the new product. You also demonstrated how generative AI can be useful for personalized product recommendations for customers.

© IBM Corporation. All rights reserved.